Chapter 7. Multivariate Time Series

VARMA models, inference, multivariate GARCH

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Portfolios and their risk measures need joint models of several series. This chapter covers multivariate stationarity, VARMA (VAR) models, cointegration, multivariate GARCH and copula-based models.

7.1 Multivariate time series models

Motivation. Stationarity, autocovariance and ARMA carry over with matrices in place of scalars; new are cross-dependence and cointegration.

$\mathbf{Y}_t = (Y_{1t}, \dots, Y_{kt})^T$ is a *k-dimensional multivariate time series*; $\mathbf{x} \le \mathbf{y}$ means $x_i \le y_i$ for all i.

Definition 7.1.1 (Strict stationarity)

A multivariate series $\{\mathbf{Y}_t\}$ is *strictly stationary* if for all integers $t_1, \dots, t_m$ and h ($m \ge 1$) and all $\mathbf{y}_1, \dots, \mathbf{y}_m \in \mathbb{R}^k$,

$$P(\mathbf{Y}_{t_1} \le \mathbf{y}_1, \dots, \mathbf{Y}_{t_m} \le \mathbf{y}_m) = P(\mathbf{Y}_{t_1+h} \le \mathbf{y}_1, \dots, \mathbf{Y}_{t_m+h} \le \mathbf{y}_m).$$

Definition 7.1.2 (Stationarity)

A multivariate series $\{\mathbf{Y}_t\}$ is *stationary* if
(1) $E\mathbf{Y}_t = \boldsymbol{\mu}$ for all t;
(2) $EY_{it}^2 < \infty$ for all $1 \le i \le k$ and all t;
(3) $\operatorname{Cov}(Y_{it}, Y_{j,t+h}) = \operatorname{Cov}(Y_{i0}, Y_{jh})$ for all $1 \le i, j \le k$ and all t, h.

Definition 7.1.3 (Autocovariance function)

For a stationary $\{\mathbf{Y}_t\}$, the $k \times k$ matrix-valued function $\Gamma(\cdot) = (\Gamma_{ij}(\cdot))_{i,j=1}^k$ with

$$\Gamma_{ij}(h) = \operatorname{Cov}(Y_{it}, Y_{j,t+h}), \qquad h = 0, \pm 1, \pm 2, \dots,$$

is the *autocovariance function*; that is, $\Gamma(h) = E(\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t+h} - \boldsymbol{\mu})^T$ with $\boldsymbol{\mu} = E\mathbf{Y}_1$.

Theorem 7.1.4 (Properties of the autocovariance function)

The autocovariance function of a stationary series satisfies:
(1) $\Gamma(-h) = \Gamma(h)^T$;
(2) $|\Gamma_{ij}(h)| \le \sqrt{\Gamma_{ii}(0)}\sqrt{\Gamma_{jj}(0)}$;
(3) for all $n \ge 1$ and $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^k$, $\sum_{i,j=1}^n \mathbf{a}_i^T \Gamma(i-j)\mathbf{a}_j \ge 0$.

Proof. [Supplement] (Book exercise.)

(1) With $s = t - h$, $\Gamma(-h) = E(\mathbf{Y}_{s+h} - \boldsymbol{\mu})(\mathbf{Y}_s - \boldsymbol{\mu})^T = \big[E(\mathbf{Y}_s - \boldsymbol{\mu})(\mathbf{Y}_{s+h} - \boldsymbol{\mu})^T\big]^T = \Gamma(h)^T$.

(2) Cauchy–Schwarz: $|\operatorname{Cov}(Y_{it}, Y_{j,t+h})| \le \sqrt{\operatorname{Var} Y_{it}}\sqrt{\operatorname{Var} Y_{j,t+h}} = \sqrt{\Gamma_{ii}(0)\Gamma_{jj}(0)}$.

(3) As $\operatorname{Cov}(\mathbf{a}_i^T\mathbf{Y}_i, \mathbf{a}_j^T\mathbf{Y}_j) = \mathbf{a}_i^T\Gamma(j-i)\mathbf{a}_j$,

$$0 \le \operatorname{Var}\Big(\sum_{i=1}^n \mathbf{a}_i^T\mathbf{Y}_i\Big) = \sum_{i,j=1}^n \mathbf{a}_i^T\Gamma(j-i)\mathbf{a}_j = \sum_{i,j=1}^n \mathbf{a}_j^T\Gamma(i-j)\mathbf{a}_i,$$

transposing each scalar and using (1); relabel $i \leftrightarrow j$. ∎

Definition 7.1.5 (Vector ARMA model)

If $\{\mathbf{Y}_t\}$ satisfies

$$\mathbf{Y}_t - \Phi_1\mathbf{Y}_{t-1} - \cdots - \Phi_p\mathbf{Y}_{t-p} = \boldsymbol{\epsilon}_t + \Theta_1\boldsymbol{\epsilon}_{t-1} + \cdots + \Theta_q\boldsymbol{\epsilon}_{t-q}, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma), \quad (7.1)$$

where $\Phi_1, \ldots, \Phi_p, \Theta_1, \ldots, \Theta_q, \Sigma$ are $k \times k$ matrices and Σ is positive definite, then $\{\mathbf{Y}_t\}$ follows a *vector ARMA(p, q) model* (VARMA(p, q)). With $q = 0$ it is a *VAR(p) model*.

With $\Phi(z) = I - \Phi_1 z - \cdots - \Phi_p z^p$, $\Theta(z) = I + \Theta_1 z + \cdots + \Theta_q z^q$, (7.1) is $\Phi(B)\mathbf{Y}_t = \Theta(B)\boldsymbol{\epsilon}_t$; causality is now a condition on the zeros of $\det \Phi(z)$.

Theorem 7.1.6 (Causal VARMA)

If

$$\det(I - \Phi_1 z - \cdots - \Phi_p z^p) \neq 0 \qquad \text{for all complex } |z| \leq 1,$$

then the VARMA model (7.1) has a stationary solution $\{\mathbf{Y}_t\}$, it is unique, and it has the representation

$$\mathbf{Y}_t = \sum_{i=0}^{\infty} C_i \boldsymbol{\epsilon}_{t-i}, \quad (7.2)$$

where $C_0 = I_k$ and the $k \times k$ matrices C_i are determined by

$$(I - \Phi_1 z - \cdots - \Phi_p z^p) \sum_{i=0}^{\infty} C_i z^i = I + \Theta_1 z + \cdots + \Theta_q z^q, \qquad |z| \leq 1. \quad (7.3)$$

With mean $\boldsymbol{\mu}$ the model reads $\mathbf{Y}_t - \boldsymbol{\mu} - \Phi_1(\mathbf{Y}_{t-1} - \boldsymbol{\mu}) - \cdots - \Phi_p(\mathbf{Y}_{t-p} - \boldsymbol{\mu}) = \boldsymbol{\epsilon}_t + \Theta_1\boldsymbol{\epsilon}_{t-1} + \cdots + \Theta_q\boldsymbol{\epsilon}_{t-q}$.

Proof. [Supplement] (Omitted in the book.) Matrix products do not commute: $\Phi(z)$ stands on the left in (7.3).

Step 1: the coefficients. $\det \Phi(z) \neq 0$ on $|z| < 1 + \delta$ for some $\delta > 0$, so $\Phi(z)^{-1} = \operatorname{adj}\Phi(z)/\det\Phi(z)$ and $C(z) := \Phi(z)^{-1}\Theta(z) = \sum_{i\geq 0} C_i z^i$ are analytic there. Convergence at $z = 1 + \delta/2$ gives $\|C_i\| \leq K(1 + \delta/2)^{-i}$, so $\sum_i \|C_i\| < \infty$, $C_0 = I$, $\Phi(z)C(z) = \Theta(z)$; likewise $\Phi(z)^{-1} = \sum_i D_i z^i$, $\sum_i \|D_i\| < \infty$.

Step 2: existence. $E\|\boldsymbol{\epsilon}_t\| \leq (\operatorname{tr}\Sigma)^{1/2}$, so (7.2) converges absolutely a.s., and in L^2 since $\left(E\|\sum_{i=m}^{n} C_i\boldsymbol{\epsilon}_{t-i}\|^2\right)^{1/2} \leq (\operatorname{tr}\Sigma)^{1/2}\sum_{i=m}^{n}\|C_i\| \to 0$. Then $E\mathbf{Y}_t = \mathbf{0}$ and

$$E\mathbf{Y}_t\mathbf{Y}_{t+h}^T = \sum_{i,j} C_i E(\boldsymbol{\epsilon}_{t-i}\boldsymbol{\epsilon}_{t+h-j}^T)C_j^T = \sum_{i=0}^{\infty} C_i \Sigma C_{i+h}^T \qquad (h \geq 0),$$

free of t. With $\Phi(z) = \sum_{j=0}^{p} A_j z^j$ ($A_0 = I$, $A_j = -\Phi_j$),

$$\Phi(B)\mathbf{Y}_t = \sum_{j=0}^{p} A_j \sum_{i\geq 0} C_i \boldsymbol{\epsilon}_{t-j-i} = \sum_{m\geq 0}\Big(\sum_{j=0}^{\min(p,m)} A_j C_{m-j}\Big)\boldsymbol{\epsilon}_{t-m} = \sum_{m=0}^{q}\Theta_m\boldsymbol{\epsilon}_{t-m},$$

the inner sum being the z^m coefficient of $\Phi(z)C(z) = \Theta(z)$.

Step 3: uniqueness. For any stationary solution, applying $\sum_i D_i B^i$ (summable, composing like power series) to $\Phi(B)\mathbf{Y}_t = \Theta(B)\boldsymbol{\epsilon}_t$ gives $\mathbf{Y}_t = \Phi(B)^{-1}\Theta(B)\boldsymbol{\epsilon}_t = \sum_i C_i\boldsymbol{\epsilon}_{t-i}$. ∎

Example 7.1.7 (Coefficients of a bivariate ARMA(1,1))

Let $k=2$, $p=q=1$, $\Phi_1=\begin{pmatrix}0.5 & 0.5\\ 0 & 0.5\end{pmatrix}$ and $\Theta_1=\Phi_1^T$. Here $\det\Phi(z)=(1-0.5z)^2\neq 0$ for $|z|\le 1$. From (7.3),

$$\sum_{i=0}^{\infty}C_iz^i=\begin{pmatrix}1-0.5z & -0.5z\\ 0 & 1-0.5z\end{pmatrix}^{-1}\begin{pmatrix}1+0.5z & 0\\ 0.5z & 1+0.5z\end{pmatrix}$$
$$=(1-0.5z)^{-2}\begin{pmatrix}1 & 0.5z(1+0.5z)\\ 0.5z(1-0.5z) & 1-0.25z^2\end{pmatrix}.$$

With $(1-0.5z)^{-2}=\sum_{m\ge0}(m+1)2^{-m}z^m$,

$$C_0=I,\qquad C_i=2^{-i}\begin{pmatrix}i+1 & 2i-1\\ 1 & 2\end{pmatrix},\quad i=1,2,\ldots$$

e.g. the $(1,2)$ entry is $\frac{1}{2}\,i\,2^{-(i-1)}+\frac{1}{4}(i-1)2^{-(i-2)}=(2i-1)2^{-i}$.

7.1.1 Cointegration

Definition 7.1.8 (Integration order and cointegration)

If $(1-B)^d\mathbf{Y}_t=\left((1-B)^dY_{1t},\ldots,(1-B)^dY_{kt}\right)^T$ is stationary while $\mathbf{Y}_t,(1-B)\mathbf{Y}_t,\ldots,(1-B)^{d-1}\mathbf{Y}_t$ are nonstationary, we write $\mathbf{Y}_t\sim I(d)$. If $\mathbf{Y}_t\sim I(d)$ and, for some positive integer $\delta<d$ and some vector $\boldsymbol{\alpha}\in\mathbb{R}^k$, $\boldsymbol{\alpha}^T\mathbf{Y}_t\sim I(\delta)$, then $\mathbf{Y}_t$ is *cointegrated* with *cointegrating vector* $\boldsymbol{\alpha}$. (The most common case is $d=1$, $\delta=0$: a stationary combination of $I(1)$ series.)

Contrast *spurious regression*: regressing one of two *independent* random walks on the other typically gives a large R^2 and a significant slope. Cointegration is a genuine long-run equilibrium.

Example 7.1.10 (Consumption and income)

Let consumption C_t and disposable income I_t (real, per capita) be $I(1)$ with $C_t=a+bI_t+u_t$, $a,b\in(0,1)$, $Eu_t=0$, $u_t\sim I(0)$. Then $\boldsymbol{\alpha}^T\mathbf{Y}_t=C_t-bI_t=a+u_t\sim I(0)$ for $\mathbf{Y}_t=(C_t,I_t)^T$, $\boldsymbol{\alpha}=(1,-b)^T$: cointegrated.

Cointegrated series are written as a *vector error correction model*:

Definition 7.1.11 (VECM)

The VECM(p) with *cointegration rank* r ($\le k$) is

$$\Delta\mathbf{Y}_t=\boldsymbol{\delta}+\Pi\mathbf{Y}_{t-1}+\sum_{i=1}^{p-1}\Phi_i^*\Delta\mathbf{Y}_{t-i}+\boldsymbol{\epsilon}_t,\qquad\{\boldsymbol{\epsilon}_t\}\sim IID(\mathbf{0},\Sigma),\tag{7.4}$$

where $\Delta=1-B$ is the differencing operator, $\boldsymbol{\delta}$ is a $k\times1$ constant vector, $\Pi=\boldsymbol{\alpha}\boldsymbol{\beta}^T$ with $\boldsymbol{\alpha},\boldsymbol{\beta}$ $k\times r$ matrices, and the Φ_i^* are $k\times k$ matrices.

If $\mathbf{Y}_t\sim I(1)$, (7.4) forces $\boldsymbol{\alpha}\boldsymbol{\beta}^T\mathbf{Y}_{t-1}$, hence ($\boldsymbol{\alpha}$ of full column rank) $\boldsymbol{\beta}^T\mathbf{Y}_{t-1}$, to be stationary: the columns of $\boldsymbol{\beta}$ are r cointegrating vectors, and $\boldsymbol{\alpha}$ gives the speed of error correction.

Johansen (1988) tests $H_0 : r \le r_0$ against $H_1 : r > r_0$ by the *trace statistic*

$$\lambda_{\text{trace}}(r_0) = -n \sum_{i=r_0+1}^{k} \log(1 - \hat{\lambda}_i), \tag{7.5}$$

with $\hat{\lambda}_1 \ge \cdots \ge \hat{\lambda}_k$ the estimated eigenvalues associated with Π; under H_0 the eigenvalues beyond r_0 vanish, so large values reject. The null law is nonstandard and depends on deterministic terms, r_0 and the dimension.

7.2 Inference for stationary multivariate series

Motivation. Sample means and autocovariance matrices are consistent, and VAR(p) is fitted by equation-wise least squares.

Theorem 7.2.1 (Sample mean)

Let $\mathbf{Y}_t = \boldsymbol{\mu} + \sum_{i=0}^{\infty} C_i \boldsymbol{\epsilon}_{t-i}$ with $\{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, \Sigma)$ and $k \times k$ matrices C_i satisfying $\sum_{i=0}^{\infty} \|C_i\| < \infty$, where $\|C\| = \sup\{\|C\mathbf{y}\| : \|\mathbf{y}\| \le 1\}$. Let $\hat{\boldsymbol{\mu}} = \frac{1}{n}\sum_{t=1}^{n} \mathbf{Y}_t$. Then:
(1) $P(\hat{\boldsymbol{\mu}} \to \boldsymbol{\mu},\ n \to \infty) = 1$;
(2) if $\Gamma_{ii}(n) \to 0$ for all $1 \le i \le k$, then $E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) \to 0$;
(3) if $\sum_{h=0}^{\infty} |\Gamma_{ii}(h)| < \infty$ for all $1 \le i \le k$, then

$$nE(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) \to \sum_{i=1}^{k} \sum_{h=-\infty}^{\infty} \Gamma_{ii}(h).$$

Proof. [Supplement] (Omitted in the book.) As in Step 2 of Theorem 7.1.6, the series converges a.s. and in L^2, and $\{\mathbf{Y}_t\}$ is stationary.

(1) $\mathbf{Y}_t = G(\boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_{t-1}, \dots)$ is strictly stationary ergodic with $E\|\mathbf{Y}_t\| < \infty$; Birkhoff gives $\hat{\boldsymbol{\mu}} \to \boldsymbol{\mu}$ a.s.

(2) $E(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}) = \sum_{i=1}^{k} \operatorname{Var}(\bar{Y}_i)$ and

$$\operatorname{Var}(\bar{Y}_i) = \frac{1}{n^2} \sum_{s,t=1}^{n} \Gamma_{ii}(t-s) = \frac{1}{n} \sum_{|h|<n} \Big(1 - \frac{|h|}{n}\Big)\Gamma_{ii}(h) \le \frac{1}{n} \sum_{|h|<n} |\Gamma_{ii}(h)|,$$

a Cesàro mean tending to 0 when $\Gamma_{ii}(h) \to 0$.

(3) $n \operatorname{Var}(\bar{Y}_i) = \sum_{|h|<n}(1 - |h|/n)\Gamma_{ii}(h) \to \sum_h \Gamma_{ii}(h)$ by dominated convergence (bound $|\Gamma_{ii}(h)|$); sum over i. ∎

Theorem 7.2.2 (Sample autocovariance)

Let $\{\mathbf{Y}_t\}$ be as in Theorem 7.2.1. Then $\Gamma(h) = \sum_{i=0}^{\infty} C_i \Sigma C_{i+h}^T$ $(h \ge 0)$, and the sample autocovariance function

$$\hat{\Gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-|h|} (\mathbf{Y}_t - \hat{\boldsymbol{\mu}})(\mathbf{Y}_{t+|h|} - \hat{\boldsymbol{\mu}})^T$$

(for $h < 0$ take the transpose) satisfies $P(\hat{\Gamma}(h) \to \Gamma(h),\ n \to \infty) = 1$; that is, $\hat{\Gamma}(\cdot)$ is a consistent estimator of $\Gamma(\cdot)$.

Proof. [Supplement] $\Gamma(h)$ is from Step 2 of Theorem 7.1.6. For $h \geq 0$, with $\mathbf{Y}_t - \hat{\boldsymbol{\mu}} = (\mathbf{Y}_t - \boldsymbol{\mu}) - (\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})$,

$$\hat{\Gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t+h} - \boldsymbol{\mu})^T - \Big(\frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_t - \boldsymbol{\mu})\Big)(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T$$
$$- (\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})\Big(\frac{1}{n}\sum_{t=1}^{n-h}(\mathbf{Y}_{t+h} - \boldsymbol{\mu})\Big)^T + \frac{n-h}{n}(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})^T.$$

$Z_t = (\mathbf{Y}_t - \boldsymbol{\mu})(\mathbf{Y}_{t+h} - \boldsymbol{\mu})^T$ is an integrable function of the shifted i.i.d. sequence, so $\frac{1}{n}\sum_{t\leq n-h} Z_t \to \Gamma(h)$ a.s.; the other terms vanish by Theorem 7.2.1(1). ■

7.2.1 Inference in VAR(p) models

For VAR(p) $\mathbf{Y}_t - \Phi_1\mathbf{Y}_{t-1} - \cdots - \Phi_p\mathbf{Y}_{t-p} = \boldsymbol{\epsilon}_t$, given $\mathbf{Y}_{-p+1}, \ldots, \mathbf{Y}_0$, let

$$\mathbf{y}_t = \begin{pmatrix} \mathbf{Y}_t \\ \vdots \\ \mathbf{Y}_{t-p+1} \end{pmatrix} (kp \times 1), \qquad \begin{aligned} Y &= (\mathbf{Y}_1, \ldots, \mathbf{Y}_n)\ (k \times n), & \Phi &= (\Phi_1, \ldots, \Phi_p)\ (k \times kp), \\ X &= (\mathbf{y}_0, \ldots, \mathbf{y}_{n-1})\ (kp \times n), & \epsilon &= (\boldsymbol{\epsilon}_1, \ldots, \boldsymbol{\epsilon}_n)\ (k \times n), \end{aligned}$$

so that $Y = \Phi X + \epsilon$.

Proposition (Least squares for VAR(p), [Supplement])

If XX^T is invertible, then for every Φ

$$(Y-\Phi X)(Y-\Phi X)^T = (Y-\hat{\Phi}X)(Y-\hat{\Phi}X)^T + (\hat{\Phi}-\Phi)XX^T(\hat{\Phi}-\Phi)^T, \qquad \hat{\Phi} = YX^T(XX^T)^{-1}.$$

Hence $\hat{\Phi}$ minimizes $\sum_{t=1}^n \|\mathbf{Y}_t - \Phi\mathbf{y}_{t-1}\|^2 = \mathrm{tr}\{(Y - \Phi X)(Y - \Phi X)^T\}$ (and the determinant of the residual cross-product matrix), uniquely.

Proof. Expand $Y - \Phi X = (Y - \hat{\Phi}X) + (\hat{\Phi} - \Phi)X$; cross terms vanish as $(Y - \hat{\Phi}X)X^T = YX^T - YX^T(XX^T)^{-1}XX^T = 0$. The second term is positive semidefinite, zero only for $\Phi = \hat{\Phi}$; take traces or determinants. ■

So

$$\hat{\Phi} = YX^T(XX^T)^{-1}, \qquad \hat{\Sigma} = \frac{1}{n - kp}(Y - \hat{\Phi}X)(Y - \hat{\Phi}X)^T,$$

and $\hat{\Phi}$ equals OLS applied to each equation.

To choose p, fit VAR(i), $i = 1, \ldots, p_0$, with residuals $\hat{\boldsymbol{\epsilon}}_t^{(i)} = \mathbf{Y}_t - \hat{\Phi}_1^{(i)}\mathbf{Y}_{t-1} - \cdots - \hat{\Phi}_i^{(i)}\mathbf{Y}_{t-i}$,

$$\hat{\Sigma}_i = \frac{1}{n - 2i}\sum_{t=i+1}^{n} \hat{\boldsymbol{\epsilon}}_t^{(i)}\hat{\boldsymbol{\epsilon}}_t^{(i)T},$$

and minimize ($k^2 i$ AR parameters)

$$AIC(i) = \log|\hat{\Sigma}_i| + 2k^2 i/n \qquad \text{or} \qquad BIC(i) = \log|\hat{\Sigma}_i| + k^2 i \log(n)/n.$$

7.3 Multivariate GARCH models

Motivation. Portfolio volatility depends on a time-varying conditional covariance matrix.

For $\mathbf{Y}_t = (y_{1t}, \dots, y_{dt})^T$, $\sigma_{ijt} = \mathrm{Cov}(y_{it}, y_{jt} \mid \mathcal{F}_{t-1})$ and $\Sigma_t = (\sigma_{ijt})$, the simplest multivariate GARCH model is

$$\mathbf{Y}_t = \Sigma_t^{1/2} \boldsymbol{\epsilon}_t, \qquad \{\boldsymbol{\epsilon}_t\} \sim IID(\mathbf{0}, I_d), \qquad E(\mathbf{Y}_t \mathbf{Y}_t^T \mid \mathcal{F}_{t-1}) = \Sigma_t, \tag{7.6}$$

with Σ_t a function of the past. An unrestricted recursion for the $d(d+1)/2$ entries has $O(d^4)$ parameters, so restricted forms are used: BEKK (Baba–Engle–Kraft–Kroner), positive definite by a quadratic-form recursion, and CCC-GARCH, $\Sigma_t = D_t R D_t$ with $D_t = \mathrm{diag}(\sigma_{11t}^{1/2}, \dots, \sigma_{ddt}^{1/2})$ from univariate GARCH and constant correlation R (Bollerslev, 1990; Engle and Kroner, 1995).

7.4 SCOMDY models

Motivation. Univariate GARCH volatilities plus a copula for the standardized errors give a flexible semiparametric multivariate model.

A simple instance of (7.6): d GARCH(1,1) series with jointly normal errors,

$$y_{it} = \sigma_{it} \xi_{it}, \quad i = 1, \dots, d, \qquad \boldsymbol{\xi}_t = (\xi_{1t}, \dots, \xi_{dt})^T \text{ i.i.d. multivariate normal.} \tag{7.7}$$

For non-normal $\boldsymbol{\xi}_t$, the *semiparametric copula-based multivariate dynamic* (SCOMDY) model uses a copula.

Theorem (Sklar, continuous case, [Supplement])

Let $\mathbf{Y} = (Y_1, \dots, Y_d)^T$ have joint distribution function F and continuous marginals $F_1, \dots, F_d$. Then there is a function C on $[0,1]^d$, namely the distribution function of $(F_1(Y_1), \dots, F_d(Y_d))$, such that

$$F(\mathbf{y}) = C(F_1(y_1), \dots, F_d(y_d)), \qquad \mathbf{y} = (y_1, \dots, y_d)^T,$$

and C has uniform $(0,1)$ marginals (a *copula*).

Proof. For $u \in (0,1)$ let $x_u = \max\{x : F_i(x) = u\}$ (exists by continuity); $\{F_i(Y_i) \le u\} = \{Y_i \le x_u\}$ a.s., so $U_i = F_i(Y_i)$ is uniform. Also $\{Y_i \le y_i\} \subset \{F_i(Y_i) \le F_i(y_i)\}$ with null difference (F_i is flat there). Hence

$$F(\mathbf{y}) = P(Y_i \le y_i \ \forall i) = P(U_i \le F_i(y_i) \ \forall i) = C(F_1(y_1), \dots, F_d(y_d)).$$

■

The copula carries all the dependence and is modelled instead of F (Gaussian, Archimedean, Frank, Clayton). For $d = 2$ the *Gaussian copula* is

$$C(u, v; \rho) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v)),$$

with Φ_ρ the standard bivariate normal d.f. with correlation ρ. An *Archimedean copula* is

$$C(u, v) = \varphi^{-1}(\varphi(u) + \varphi(v))$$

for a convex decreasing generator φ on $[0,1]$, $\varphi(1) = 0$: $\varphi(t) = \log(e^{-\theta} - 1) - \log(e^{-\theta t} - 1)$, $\theta \neq 0$ (*Frank*), $\varphi(t) = (t^{-\theta} - 1)/\theta$, $\theta > 0$ (*Clayton*); see Joe (1997).

Marginals are estimated empirically: $\hat{F}(\mathbf{y}) = C(\hat{F}_1(y_1), \dots, \hat{F}_d(y_d))$ with

$$\hat{F}_i(y) = \frac{1}{n} \sum_{j=1}^{n} I(Y_{ij} \le y), \qquad i = 1, \dots, d,$$

giving a conditional likelihood for (7.7) with copula errors. SCOMDY contains GARCH, VAR and Markov-switching models; Chen and Fan (2006) estimate the copula parameters by a Kullback–Leibler criterion, with asymptotics and model-selection tests.

7.5 Selected exercises

Motivation. Exercises on geometric decay of VARMA autocovariances and the ACVF of a bivariate ARMA(1,1).

Exercise 7.2 (Geometric decay of the autocovariances)

Let $\{\mathbf{Y}_t\}$ be the stationary solution of a VARMA model satisfying the condition of Theorem 7.1.6. Show that there are $\rho \in (0,1)$ and a constant K such that $|\Gamma_{ij}(h)| \le K\rho^{|h|}$ for all i, j and h.

Solution.

Step 1: decay of C_i. By Step 1 of Theorem 7.1.6, $\|C_i\| \le K_0\rho^i$ with $\rho = (1+\delta/2)^{-1} \in (0,1)$.

Step 2: bound on $\Gamma(h)$. By Theorem 7.2.2, for $h \ge 0$,

$$\|\Gamma(h)\| \le \sum_{i\ge 0}\|C_i\|\,\|\Sigma\|\,\|C_{i+h}\| \le \|\Sigma\|K_0^2\sum_{i\ge 0}\rho^i\rho^{i+h} = \frac{\|\Sigma\|K_0^2}{1-\rho^2}\,\rho^h.$$

Step 3: entries. $|\Gamma_{ij}(h)| = |\mathbf{e}_i^T\Gamma(h)\mathbf{e}_j| \le \|\Gamma(h)\|$, and $\Gamma(h) = \Gamma(-h)^T$ for $h<0$; so $K = \|\Sigma\|K_0^2/(1-\rho^2)$. ■

Exercise 7.3 (Autocovariances of a bivariate ARMA(1,1))

Find the autocovariance matrix function of the bivariate ARMA(1,1) model $\mathbf{Y}_t = \Phi\mathbf{Y}_{t-1} + \boldsymbol{\epsilon}_t + \Theta\boldsymbol{\epsilon}_{t-1}$ with

$$\Phi = \begin{pmatrix} 0.7 & 0 \\ 0 & 0.6 \end{pmatrix}, \qquad \Theta = \begin{pmatrix} 0.5 & 0.6 \\ -0.7 & 0.8 \end{pmatrix}, \qquad \{\boldsymbol{\epsilon}_t\} \sim \text{IID } N\Big(\mathbf{0}, \begin{pmatrix} 1 & 0.71 \\ 0.71 & 2 \end{pmatrix}\Big).$$

Solution.

Step 1: causality. $\det(I-\Phi z) = (1-0.7z)(1-0.6z)$ has zeros $1/0.7$, $1/0.6$; by Theorem 7.1.6, $\mathbf{Y}_t = \sum_{i\ge 0} C_i\boldsymbol{\epsilon}_{t-i}$, so $E\mathbf{Y}_t = \mathbf{0}$, $E\mathbf{Y}_t\boldsymbol{\epsilon}_{t+j}^T = 0$ $(j \ge 1)$, $E\mathbf{Y}_t\boldsymbol{\epsilon}_t^T = \Sigma$.

Step 2: lag 0. With $\mathbf{U}_t = \boldsymbol{\epsilon}_t + \Theta\boldsymbol{\epsilon}_{t-1}$, $E\mathbf{Y}_{t-1}\mathbf{U}_t^T = \Sigma\Theta^T$, $E\mathbf{U}_t\mathbf{U}_t^T = \Sigma + \Theta\Sigma\Theta^T$, so

$$\Gamma(0) = \Phi\Gamma(0)\Phi^T + Q, \qquad Q = \Phi\Sigma\Theta^T + \Theta\Sigma\Phi^T + \Sigma + \Theta\Sigma\Theta^T.$$

Numerically $\Sigma\Theta^T = \begin{pmatrix} 0.926 & -0.132 \\ 1.555 & 1.103 \end{pmatrix}$, $\Phi\Sigma\Theta^T = \begin{pmatrix} 0.6482 & -0.0924 \\ 0.9330 & 0.6618 \end{pmatrix}$ ($\Theta\Sigma\Phi^T$ its transpose), $\Theta\Sigma\Theta^T = \begin{pmatrix} 1.3960 & 0.5958 \\ 0.5958 & 0.9748 \end{pmatrix}$, hence

$$Q = \begin{pmatrix} 3.6924 & 2.1464 \\ 2.1464 & 4.2984 \end{pmatrix}.$$

As Φ is diagonal, $\Gamma_{ij}(0) = Q_{ij}/(1-\phi_i\phi_j)$:

$$\Gamma(0) = \begin{pmatrix} 3.6924/0.51 & 2.1464/0.58 \\ 2.1464/0.58 & 4.2984/0.64 \end{pmatrix} = \begin{pmatrix} 7.2400 & 3.7007 \\ 3.7007 & 6.7163 \end{pmatrix}.$$

Step 3: lag 1 and beyond. For $h \ge 1$,

$$\Gamma(h) = E\mathbf{Y}_t\big(\Phi\mathbf{Y}_{t+h-1} + \boldsymbol{\epsilon}_{t+h} + \Theta\boldsymbol{\epsilon}_{t+h-1}\big)^T = \Gamma(h-1)\Phi^T + E(\mathbf{Y}_t\boldsymbol{\epsilon}_{t+h-1}^T)\Theta^T,$$

the last term being $\Sigma\Theta^T$ for $h = 1$ and 0 for $h \geq 2$:

$$\Gamma(1) = \Gamma(0)\Phi^T + \Sigma\Theta^T = \begin{pmatrix} 5.9940 & 2.0884 \\ 4.1455 & 5.1328 \end{pmatrix}, \qquad \Gamma(h) = \Gamma(1)(\Phi^T)^{h-1} \ (h \geq 1),$$

i.e. $\Gamma_{i1}(h) = 0.7^{h-1}\Gamma_{i1}(1)$, $\Gamma_{i2}(h) = 0.6^{h-1}\Gamma_{i2}(1)$, and $\Gamma(-h) = \Gamma(h)^T$. The contemporaneous correlation is $3.7007/\sqrt{7.24 \cdot 6.7163} \approx 0.531$; $\Gamma(1)$ is not symmetric (4.1455 vs 2.0884), mainly through $(\Sigma\Theta^T)_{21} = 1.555$ vs $(\Sigma\Theta^T)_{12} = -0.132$. ■